

Ming-Hung Fan

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EXPERIENCE

Collaborate Technology | Frontend Engineer | Taipei City, Taiwan Aug 2022 – Aug 2024

- Architected a legacy system rebuild and refactor using **Angular 18** and **JWT** authentication, increasing stability by **20%**
- Deployed a cross-platform **Android** application using **Ionic**, improving **SEO** by **10 points**
- Mentored junior developers while developing custom **TypeScript** automation scripts to improve development velocity by **6 hours/week** and maintain comprehensive **technical documentation**

Allion Labs | Frontend Web Developer | Nantou, Taiwan Feb 2022 – Jul 2022

- Developed responsive **Angular** web applications that integrated secure **RESTful APIs** for real-time data synchronization
- Developed a modular UI system using **TailwindCSS** and **Angular Material**, boosting cross-team feature delivery by **25%**
- Created custom **IoT widgets** and automated backend rule chains on **ThingsBoard** for real-time telemetry visualization of **50+** simultaneous device streams

Knowtecq Limited | Operations Engineer | Taipei City, Taiwan Jan 2019 – Jul 2021

- Maintained **99.9%** system uptime by configuring complex routing and firewall policies across **Cisco**, **Juniper**, **Palo Alto**, and **Fortinet** infrastructures while engineering **Python** automation scripts for traceroute parsing and IP data retrieval on **Netflow** and **Noction** platforms to expedite incident response
- Successfully mitigated client **DDoS** attacks via dynamic traffic redirection using deep traffic analysis with **Grafana** and **Zabbix**

Republic of China Military Police | Information and Network Officer | Taiwan Jul 2008 – Jul 2018

- Led and managed cross-functional technical units, from platoon to company-sized teams of **100+**, managing large-scale **IT budgets**, hardware maintenance contracts, and infrastructure renovation projects

PROJECTS

Sir Isaac's Vocab Studio | React 19, Google Gemini 2.5 Flash, IndexedDB | [\[Website\]](#)

- Developed an interactive **AI** learning environment, utilizing **React 19** and **Google Gemini API** for automatic vocabulary generation, incorporating **IndexedDB** for caching **50MB+** and efficient UI state management for high-resolution **PDF/PNG exports**

Casa Sueño del Mar | React 19, Vite 6, Supabase, Tailwind CSS v4 | [\[Website\]](#)

- Created a bilingual vacation rental web application with **React 19**, **TypeScript**, and **Supabase (PostgreSQL & Storage)**, including a custom **CMS** with drag-and-drop asset management (**@dnd-kit**) to make content updates easier

Air Quality Monitoring Platform | Angular, Tailwind CSS, WebSocket, JWT, ApexCharts | [\[Website\]](#)

- Developed a real-time environment monitoring dashboard with live data streaming using **WebSockets** and **RESTful APIs**, and implemented automated **JWT** token refresh capabilities to ensure session security and data consistency

ShengShi | React, Redux, Firebase, Algolia | [\[Website\]](#)

- Developed a **React** food-sharing platform using **Redux** for state management, integrating the **Google Maps API** to display restaurant locations and **QR code** verification for secure exchanges
- Implemented a serverless architecture using **Firebase**, integrating **Algolia** for real-time search, **Firebase Auth** for secure login, and **Cypress** for **E2E testing**

TECHNICAL SKILLS

- **Languages:** TypeScript, JavaScript (ES6+), Python, C, C++, C#, HTML5, CSS3, SQL
- **Frameworks & Libraries:** React, Angular, Node.js, Express.js, Ionic, TailwindCSS, Bootstrap, Redux, Jest, .NET
- **AI & Cloud Infrastructure:** Prompt Engineering, Google Gemini API, Generative AI Architecture, AWS
- **Tools & Developer Platforms:** Git, Docker, CI/CD, Jenkins, Cisco Routing & Switching (CCNP), Grafana, Zabbix, PostgreSQL, MongoDB, Firebase, SDL
- **Professional Certifications:** CCNP, ITIL 4 Foundation, PCAP Python, Scrum Fundamentals Certified

EDUCATION & TRAINING

Seneca Polytechnic | Diploma in Computer Programming | GPA: 4.0 / 4.0 | Toronto, ON Apr 2026

National Defense University | Bachelor of Electrical and Electronic Engineering | Taiwan Jul 2008

AiWorks School & III Training | Advanced Software & Network Engineering Bootcamps | Taiwan 2018, 2021